

JAL2301 DATA SCIENCE

UNIT – II

DESCRIPTIVE STATISTICS - DESCRIBING DATA

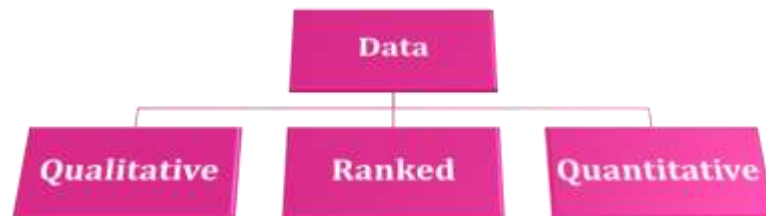
UNIT – II DESCRIPTIVE STATISTICS - DESCRIBING DATA SYLLABUS

Types of Data - Types of Variables - Describing Data with Tables and Graphs - Describing Data with Averages - Describing Variability - Normal Distributions and Standard (z) Scores.

Types of Data

TYPES OF DATA

- Any statistical analysis is performed on data.
- Data → a collection of actual observations or scores in a survey or an experiment.
- The precise form of a statistical analysis often depends on the type of data.



TYPES OF DATA

1. Qualitative data → A set of observations where any single observation is a word, letter, or numerical code that represents a class or category.

- Consist of words (Yes or No, Male or Female), letters (Y or N, (M or F), or numerical codes (0 or 1) that represent a class or category.

2. Ranked data → A set of observations where any single observation is a number that indicates relative standing

- Consist of numbers (1st, 2nd, ... 40th place) that represent relative standing within a group.

3. Quantitative data → set of observations where any single observation is a number that represents an amount or a count.

- Consist of numbers ((weights (238kg), height of a person (160 cm), number of participants (85)) that represent an amount or a count.

TYPES OF DATA

Eg.

NAME	GENDER	SCORE	POSITION
Prabhu	Male	85	2
Shalini	Female	78	3
Meera	Female	43	5
Nizam	Male	92	1
Joshua	Male	57	4

↓
Qualitative
data

↓
Quantitative
data

↓
Ranked
data

TYPES OF DATA

- Indicate whether each of the following terms is *qualitative*, *ranked* or *quantitative*.

No.	DATA CATEGORY	Type of Data
1.	Ethnic Group	
2.	Age	
3.	Family Size	
4.	Academic Major	
5.	Gender Preference	

No.	DATA CATEGORY	Type of Data
6.	IQ Score	
7.	Net Worth (Dollars)	
8.	Third-place Finish	
9.	Gender	
10.	Temperature	

TYPES OF DATA

- Indicate whether each of the following terms is *qualitative*, *ranked* or *quantitative*.

No.	DATA CATEGORY	Type of Data
1.	Ethnic Group	Qualitative
2.	Age	Quantitative
3.	Family Size	Quantitative
4.	Academic Major	Qualitative
5.	Gender Preference	Qualitative

No.	DATA CATEGORY	Type of Data
6.	IQ Score	Quantitative
7.	Net Worth (Dollars)	Quantitative
8.	Third-place Finish	Ranked
9.	Gender	Qualitative
10.	Temperature	Quantitative

Types of Variables

Types of Variables

- Quantitative Variables
 - Discrete Variable
 - Continuous Variable
- Independent Variables
- Dependent Variables
- Confounding Variable

TYPES OF VARIABLES

- A **variable** → a characteristic or property that can take on different values.
- A variable can be a quantitative variable or qualitative variable.
- Quantitative variables can be discrete or continuous.
- A **discrete variable** → consists of **isolated** numbers separated by gaps. Examples → most counts: such as the number of children in a family, Indian population.
- A **continuous variable** → consists of numbers whose values have no restrictions. Examples → amounts, such as weights of male students; durations, such as the reaction times of grade school children to a fire alarm; and standardized test scores, such as those on the NEET.

Approximate Numbers

- In theory, values for continuous variables can be carried out infinitely far.
- Whenever values are rounded off, the resulting numbers are **approximate**, never exact. It is always the case with actual values for continuous variables.
- Eg. A student whose weight is listed as 70 kg could actually weigh between 69.5 and 70.5 kg.
- Eg.
 - (a) cooking time for noodles → continuous
 - (b) IQ → continuous
 - (c) age → continuous
 - (d) population of your hometown → discrete
 - (e) speed of a jetliner → continuous

Independent and Dependent Variables

- The presence or absence of a relationship between two (or more) variables.
- For example, an educationist might wish to investigate whether students who undergo special tuitions in mathematics tend to have score less than those who do not attend any tuitions → to check the relationship between tuition & score.
- Tuition → independent variable
- Score → dependent variable → When a variable is believed to have been influenced by the independent variable, it is called a dependent variable → values cannot be specified in advance

TUITION

EXAM SCORE

Independent or Dependent Variable

- You should be able to identify these two types of variables.
- In an experiment,
 - What is being **manipulated** by the investigator at the outset → qualifies as the independent variable.
 - What is measured, counted, or recorded by the investigator at the **completion** of the study → qualifies as the dependent variable.
- Once these two variables have been identified, they can be used to describe the problem posed by the study - that is, **does the independent variable cause a change in the dependent variable?**

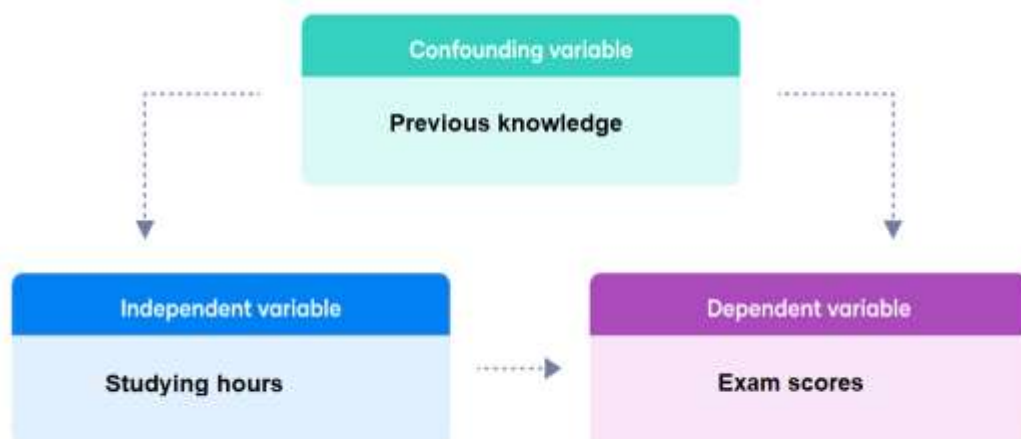
Observational Studies

- Instead of undertaking an experiment, an investigator might simply observe the relation between two variables.
- Eg. collecting measures of poverty level and crime rate for each individual in some group
 - If any statistical analysis reveals that they are correlated, they can be predicted by each other.
 - Or they were caused by some other factors.
- An **observational study** focuses on detecting relationships between variables **not manipulated** by the investigator, and it yields less clear-cut conclusions about cause-effect relationships than does an experiment.

Confounding Variable

- Whenever groups differ not just because of the independent variable but also because some **uncontrolled variable** co-varies with the independent variable, any conclusion about a cause-effect relationship is suspect.
- It is an **unmeasured third variable** that influences both the supposed cause and the supposed effect.
- If A causes B, then C is a confounding variable that can change B and A.
- The confounding variables are far harder to spot.

Confounding Variable



Confounding Variable

DEFINITION

A confounding variable is an external factor in an experiment that affects both the independent and dependent variables. It can introduce error or bias, making it difficult to determine if the observed effects are solely due to the independent variable. Controlling for them is essential to establish true causal relationships.

EXAMPLE

A study that aims to determine the effect of studying hours on exam scores may not take into account the participants' previous knowledge on the subject. In this case, previous knowledge serves as a confounding variable, as it can impact both the amount of time spent studying and the exam scores achieved.

Describing Data with Tables and Graphs

Describing Data with Tables

- Frequency Distributions For Quantitative Data
- Guidelines for Frequency Distributions
- Real Limits of Class Intervals
- Constructing Frequency Distributions
- Outliers
- Relative Frequency Distributions
- Cumulative Frequency Distributions
- Frequency Distributions for Qualitative Data

- **Frequency Distributions For Quantitative Data** Help us to detect any pattern in the data (assuming a pattern exists) by superimposing some order on the inevitable (expected) variability among observations.
- **Graphs For Quantitative Data** Graphs of frequency distributions further aid our effort to detect data patterns and make sense out of the data.
- Statistics supplies some tools, including tables and graphs, and some guidelines – to get a well-summarized set of data.

Frequency Distributions For Quantitative Data

- A frequency distribution is a collection of observations produced by sorting (arranging) observations into classes (categories) and showing their frequency (f) of occurrence in each class
- Helps us to detect any pattern in the data → frequency distribution for ungrouped data.

Frequency Distributions For Quantitative Data

TABLE 1.1
QUANTITATIVE DATA: WEIGHTS (IN POUNDS) OF MALE
STATISTICS STUDENTS

160	168	133	170	150	165	158	165
193	169	245	160	152	190	179	157
226	160	170	180	150	156	190	156
157	163	152	158	225	135	165	135
180	172	160	170	145	185	152	
205	151	220	166	152	159	156	
165	157	190	206	172	175	154	

FREQUENCY
DISTRIBUTION
(UNGROUPED DATA)

WEIGHT	f
245	1
244	0
243	0
242	0
*	
*	
*	
161	0
160	4
159	1
158	2
157	3
*	
*	
*	
136	0
135	2
134	0
133	1
Total	53

Ungrouped Data

- Frequency distributions for ungrouped data are much more informative → when the number of possible values is less than about 20. (but 133 to 245 has more than 100 possible values).

Frequency Distributions For Quantitative Data

TABLE 1.1
QUANTITATIVE DATA: WEIGHTS (IN POUNDS) OF MALE
STATISTICS STUDENTS

160	168	133	170	150	165	158	165
193	169	245	160	152	190	179	157
226	160	170	180	150	156	190	156
157	163	152	158	225	135	165	135
180	172	160	170	145	185	152	
205	151	220	166	152	159	156	
165	157	190	206	172	175	154	

FREQUENCY
DISTRIBUTION
(GROUPED DATA)

WEIGHT	f
240–249	1
230–239	0
220–229	3
210–219	0
200–209	2
190–199	4
180–189	3
170–179	7
160–169	12
150–159	17
140–149	1
130–139	3
Total	53

Grouped Data

- When the number of possible values is more than 20 → use frequency distribution for grouped data.
- Data are grouped into class intervals with 10 possible values each.



Frequency Distributions For Quantitative Data

TABLE 1.1
QUANTITATIVE DATA: WEIGHTS (IN POUNDS) OF MALE
STATISTICS STUDENTS

160	168	133	170	150	165	158	165
193	169	245	160	152	190	179	157
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180	172	160	170	145	185	152	
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FREQUENCY
DISTRIBUTION
(GROUPED DATA)

WEIGHT	<i>f</i>
240–249	1
230–239	0
220–229	3
210–219	0
200–209	2
190–199	4
180–189	3
170–179	7
160–169	12
150–159	17
140–149	1
130–139	3
Total	53

Grouped Data - Observations

- Although ranging from the 130s to the 240s, the weights peak in the 150s, with a progressively decreasing but relatively heavy concentration in the 160s and 170s.
- The distribution of weights is not balanced about its peak, but tilted in the direction of the heavier weights.

Guidelines for Frequency Distributions

S.No.	GUIDELINES	IMPORTANCE
1.	Each observation should be included in one, and only one, class.	Essential
2.	List all classes, even those with zero frequencies.	
3.	All classes should have equal intervals.	
4.	All classes should have both an upper boundary and a lower boundary.	Optional
5.	Select the class interval from convenient numbers, such as 1, 2, 3, . . . 10, particularly 5 and 10 or multiples of 5 and 10.	
6.	The lower boundary of each class interval should be a multiple of the class interval.	
7.	Aim for a total of approximately 10 classes	

Guidelines for Frequency Distributions

How Many Classes?

- 7th guideline;
- too many classes (130-134, 135-139, ..., 245-249) → provide a reasonably concise description of data.
- too few classes (100-149, 150-199, 200-249) → can mask important data patterns.

When There Are Either Many or Few Observations

- Roughly 10 number of classes

Gaps between Classes

- The size of the gap should always equal one unit of measurement (... 180-189, 190-199, ...) - the smallest possible difference between scores within a particular set of data.
- If the values are like 160.5, 181.8, etc., the class intervals can be like ...180.0-189.9, 190.0-199.9, ... and the size of the gaps can be 0.1

Real Limits of Class Intervals

- The real limits are located at the midpoint of the gap between adjacent tabled boundaries
- → that is, one-half of one unit of measurement below the lower tabled boundary
- → and one-half of one unit of measurement above the upper tabled boundary.
- Eg. the real limits for 140–149 are 139.5 and 149.5
- To avoid missing any observations.

Constructing Frequency Distributions

Step-by-step procedure

1. Find the range → the difference between the largest and smallest observations.
2. Find the class interval required to span the range by dividing the range by the desired number of classes (ordinarily 10).
3. Round off to the nearest convenient interval (such as 1, 2, 3, . . . 10, particularly 5 or 10 or multiples of 5 or 10).
4. Determine where the lowest class should begin.
5. Determine where the lowest class should end → by adding the class interval to the lower boundary and then subtracting one unit of measurement.
6. Working upward, list as many equivalent classes as are required to include the largest observation
7. Indicate with a tally the class in which each observation falls.
8. Replace the tally count for each class with a number - the frequency (f) - and show the total of all frequencies.
9. Supply headings for both columns and a title for the table

Constructing Frequency Distributions

Step-by-step procedure

1. **Range** = largest observation - smallest observation = $245 - 133 = 112$
2. **Class interval** = $\frac{\text{range}}{\text{desired number of classes}} = \frac{112}{10} = 11.2$
3. 11.2 rounded off to 10
4. Smallest observation = 133 $\rightarrow$ lowest class = 130
5. End of lowest class = $130 + 10 - 1 = 139$
6. 130-139 to 240-249
7. Tally eg.

Type of smartphone	Tally	Frequency
Android		12
Window Phone		8
Iphone		15
Amazon's fire phone		5
		Sum = 40

Frequency distribution of types of smartphones owned

Outliers

- Very extreme scores, or outliers
- Eg. A GPA of 0.06, an IQ of 170 (common level 85 to 115)

Check for Accuracy

- Was a correct GPA of 3.06 recorded erroneously as 0.06 $\rightarrow$ If the outlier survives an accuracy check, it should be treated as a legitimate score.

Might Exclude from Summaries

- Segregate an outlier from any summary of the data.
- Transfer it to a footnote instead of using excessively wide class intervals

Might Enhance Understanding

- Since a valid outlier can be viewed as the product of special circumstances, it might help you to understand the data. (eg. extremely low (or high) crime rate)

Outliers

- **Eg.** Identify any outliers in each of the following sets of data collected from nine college students.

SUMMER INCOME	AGE	FAMILY SIZE	GPA
\$6,450	20	2	2.30
\$4,820	19	4	4.00
\$5,650	61	3	3.56
\$1,720	32	6	2.89
\$600	19	18	2.15
\$0	22	2	3.01
\$3,482	23	6	3.09
\$25,700	27	3	3.50
\$8,548	21	4	3.20

- Outliers: Summer income - \$25,700; Age - 61;
Family size - 18; GPA - Nil

Relative Frequency Distributions

- Relative frequency distributions show the frequency of each class as a part or fraction of the total frequency for the entire distribution.
- Allows us to focus on the relative concentration of observations among different classes within the same distribution.
- Eg. weight in the interval 160-169 is about $\frac{12}{53} \times 100 = 23\%$ of the entire 53 observations.
- Especially helpful when two or more distributions must be compared based on different total numbers of observations.

FREQUENCY DISTRIBUTION (GROUPED DATA)

WEIGHT	<i>f</i>
240–249	1
230–239	0
220–229	3
210–219	0
200–209	2
190–199	4
180–189	3
170–179	7
160–169	12
150–159	17
140–149	1
130–139	3
Total	53

Relative Frequency Distributions

Constructing Relative Frequency Distributions

- To convert a frequency distribution into a relative frequency distribution, divide the frequency for each class by the total frequency for the entire distribution.
- Eg. weight in the interval 150-159 is about $\frac{17}{53} = 0.32$ of the entire 53 observations $\rightarrow$ proportion.
- To use relative frequency as %, multiply proportion by 100.

RELATIVE FREQUENCY DISTRIBUTION		
WEIGHT	<i>f</i>	RELATIVE <i>f</i>
240-249	1	.02
230-239	0	.00
220-229	3	.06
210-219	0	.00
200-209	2	.04
190-199	4	.08
180-189	3	.06
170-179	7	.13
160-169	12	.23
150-159	17	.32
140-149	1	.02
130-139	3	.06
Total	53	1.02*

Relative Frequency Distributions

Proportions or percentages?

- A proportion always varies between 0 and 1 $\rightarrow$ sum of the relative frequencies of all classes equals 1.
- A percentage always varies between 0 percent and 100 percent – no decimal point – mostly preferred.

RELATIVE FREQUENCY DISTRIBUTION		
WEIGHT	<i>f</i>	RELATIVE <i>f</i>
240-249	1	.02
230-239	0	.00
220-229	3	.06
210-219	0	.00
200-209	2	.04
190-199	4	.08
180-189	3	.06
170-179	7	.13
160-169	12	.23
150-159	17	.32
140-149	1	.02
130-139	3	.06
Total	53	1.02*

Cumulative Frequency Distributions

- Cumulative frequency distributions show the total number of observations in each class and in all lower-ranked classes.
- Used effectively with sets of scores, (eg. test scores for aptitude tests), when relative standing within the distribution assumes primary importance.
- Cumulative frequencies can be converted into cumulative percentages → called as **percentile ranks**.

CUMULATIVE FREQUENCY DISTRIBUTION			
WEIGHT	f	CUMULATIVE f	CUMULATIVE PERCENT
240-249	1	53	100
230-239	0	52	98
220-229	3	52	98
210-219	0	49	92
200-209	2	49	92
190-199	4	47	89
180-189	3	43	81
170-179	7	40	75
160-169	12	33	62
150-159	17	21	40
140-149	1	4	8
130-139	3	3	6
Total	53		

Cumulative Frequency Distributions

Constructing Cumulative Frequency Distributions

- To convert a frequency distribution into a cumulative frequency distribution, add to the frequency of each class the sum of the frequencies of all classes ranked below it.
- Begin with the lowest-ranked class in the frequency distribution and work upward, finding the cumulative frequencies in ascending order.

Cumulative Percentages

- If relative standing within a distribution is particularly important, then cumulative frequencies are converted to cumulative percentages.

CUMULATIVE FREQUENCY DISTRIBUTION			
WEIGHT	f	CUMULATIVE f	CUMULATIVE PERCENT
240-249	1	53	100
230-239	0	52	98
220-229	3	52	98
210-219	0	49	92
200-209	2	49	92
190-199	4	47	89
180-189	3	43	81
170-179	7	40	75
160-169	12	33	62
150-159	17	21	40
140-149	1	4	8
130-139	3	3	6
Total	53		

- For the class 150 – 159, cumulative percentage = $\frac{\text{cumulative } f}{53} \times 100$

$$= \frac{21}{53} \times 100 = 40\%$$

Cumulative Frequency Distributions

Percentile Ranks

- The percentile rank of a score indicates the **percentage of scores** in the entire distribution with similar or smaller values than that score.
- Cumulative percentages → referred to as percentile ranks.
- To describe the relative position of any score within its parent distribution

Exact and Approximate Percentile Ranks

- Exact percentile ranks → when cumulative percentages are obtained from frequency distributions for ungrouped data.
- Approximate percentile ranks → when cumulative percentages are obtained from frequency distributions for grouped data

Frequency Distributions for Qualitative Data

- Qualitative data (nominal data) → the data in any single observation as a word, letter, or numerical code among a set of observations
- Frequency distributions for qualitative data → easy to construct.
- Determine the frequency with which observations occupy each class, and report these frequencies. Eg..

Ordered Qualitative data

- When qualitative data have an ordinal level of measurement, they can be listed in descending order. Eg..

FACEBOOK PROFILE SURVEY	
Response	f
Yes	56
No	27
Total	83

Cadre	f
Principal	20
Professor	45
Associate Professor	180
Assistant Professor	300
Total	545

Frequency Distributions for Qualitative Data

Relative and Cumulative Distributions for Qualitative Data

- Relative frequency distributions and cumulative frequencies and cumulative percentages (for ordinal measurements) can be used for qualitative variables.

Cadre	f	Relative Frequency as Proportion	Relative Frequency as Percentage	Cumulative Frequency	Cumulative Percent
Principal	20	0.04	3.67	545	100.0
Professor	45	0.08	8.26	525	96.3
Associate Professor	180	0.33	33.03	480	88.1
Assistant Professor	300	0.55	55.05	300	55.0
Total	545	1.00	100.01		

Frequency Distributions

Important definitions to study:

- Frequency distribution
- Range in frequency distribution
- Real limits
- Outliers
- Relative frequency distributions
- Cumulative frequency distributions
- Percentile Ranks

Describing Data with Tables

- Frequency Distributions For Quantitative Data
- Guidelines for Frequency Distributions
- Real Limits of Class Intervals
- Constructing Frequency Distributions
- Outliers
- Relative Frequency Distributions
- Cumulative Frequency Distributions
- Frequency Distributions for Qualitative Data

Describing Data with Tables and Graphs

Describing Data with Graphs

- Graphs For Quantitative Data
- Histograms
- Frequency Polygons
- Stem and Leaf Displays
- Typical Shapes
- A Graph For Qualitative (Nominal) Data

Graphs

- Data can be described clearly and concisely with the help of a well-constructed frequency distribution.
- And data can often be described even more brilliantly / colorfully, by converting **frequency distributions into graphs** → helpful when it has to be communicated with a general audience.
- Few types of graphs:
 - Histograms
 - Frequency polygon
 - Stem and Leaf Displays

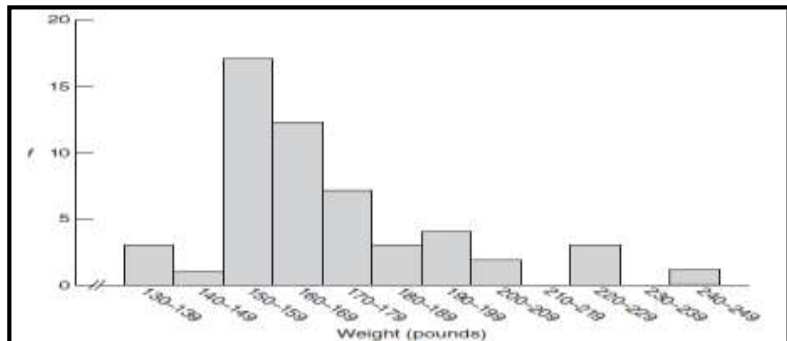
Graphs for Quantitative Data

1. Histograms

- A diagram consisting of rectangles whose area is proportional to the frequency of a variable and whose **width is equal to the class interval**.

**FREQUENCY
DISTRIBUTION
(GROUPED DATA)**

WEIGHT	<i>f</i>
240-249	1
230-239	0
220-229	3
210-219	0
200-209	2
190-199	4
180-189	3
170-179	7
160-169	12
150-159	17
140-149	1
130-139	3
Total	53



Observation from histogram: A dense concentration of weights among the 150s, 160s, and 170s, with a spread in the direction of the heavier weights.

Graphs for Quantitative Data

Histograms - important features of histograms

- Equal units along the **horizontal axis** (the X axis, or abscissa) reflect the various class intervals of the frequency distribution.
- Equal units along the **vertical axis** (the Y axis, or ordinate) reflect increases in frequency. (The units along the vertical axis do not have to be the same width as those along the horizontal axis.)
- The **intersection** of the two axes defines the origin at which both numerical scales equal 0.
- Numerical scales always increase from left to right along the horizontal axis and from bottom to top along the vertical axis.
- The body of the histogram consists of a **series of bars** whose heights reflect the frequencies for the various classes.

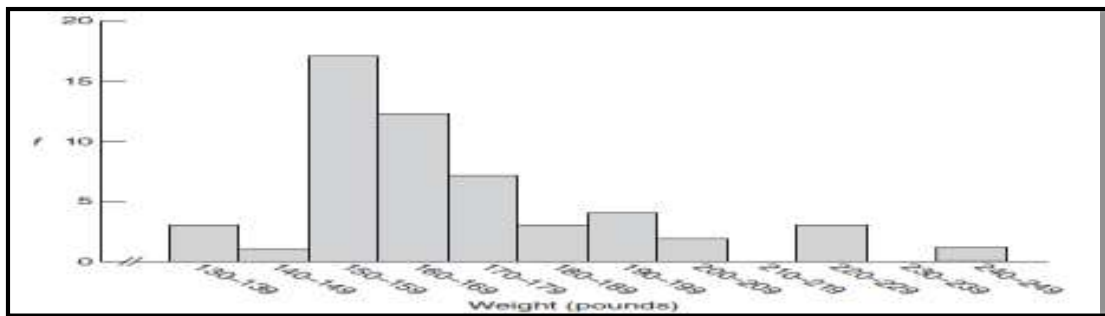
Graphs for Quantitative Data

2. Frequency Polygon

- Frequency polygon or line graph → An important variation on a histogram.
- Frequency polygons may be constructed directly from frequency distributions.

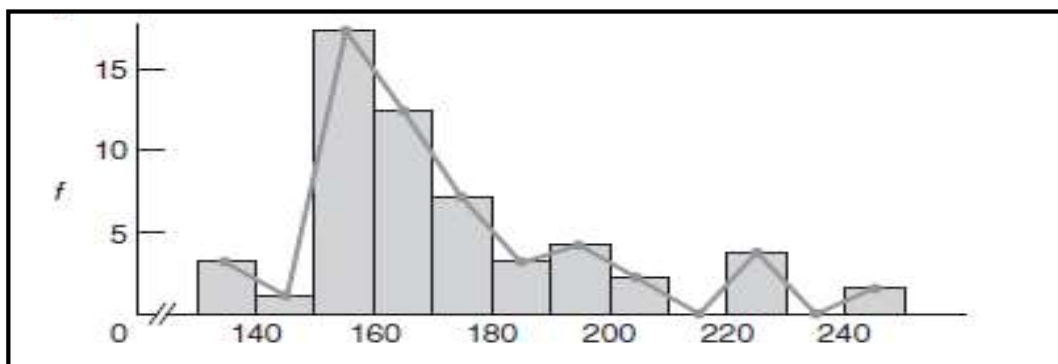
Step-by-step transformation of a histogram into a frequency polygon:

Step 1: Start with the histogram for the weight distribution



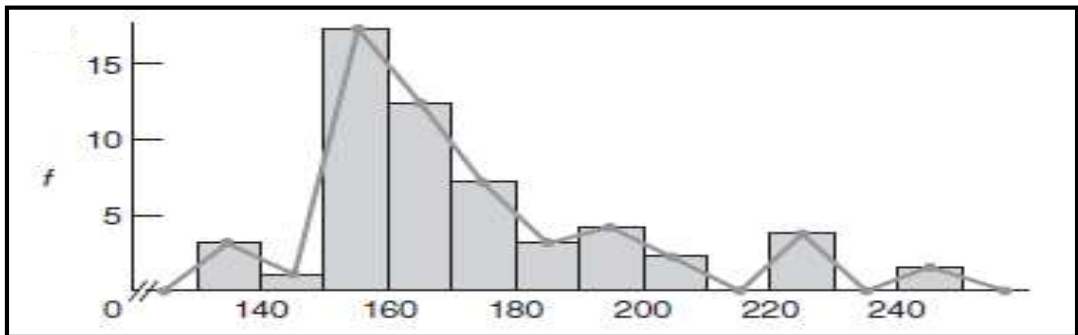
Graphs for Quantitative Data - Frequency Polygon

- Step 2: Place dots at the midpoints of each bar top or, in the absence of bar tops, at midpoints for classes on the horizontal axis, and connect them with straight lines. For the class 160-169, the midpoint = $(160+169) / 2 = 164.5$



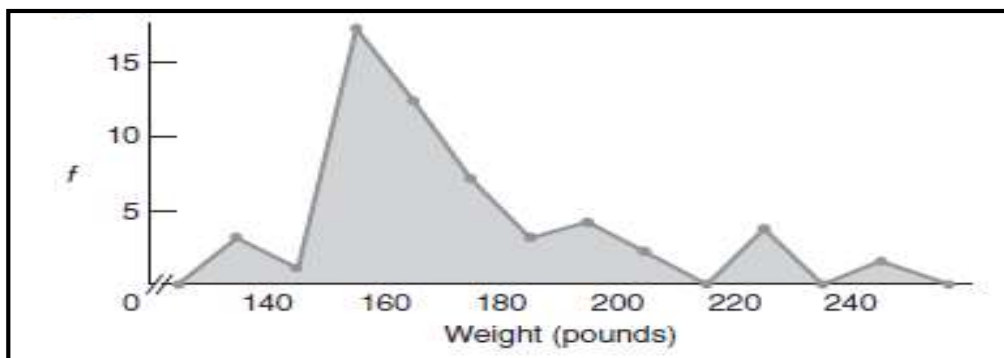
Graphs for Quantitative Data - Frequency Polygon

- Step 3: Anchor the frequency polygon to the horizontal axis. First, extend the upper tail to the midpoint of the first unoccupied class (250–259) on the upper flank of the histogram. Then extend the lower tail to the midpoint of the first unoccupied class (120–129) on the lower flank of the histogram. Now all of the area under the frequency polygon is enclosed completely.



Graphs for Quantitative Data - Frequency Polygon

- Step 4: Erase all of the histogram bars, leaving only the frequency polygon.



Graphs for Quantitative Data



3. Stem and Leaf Displays

- A device for sorting quantitative data on the basis of leading and trailing digits.
- Ideal for summarizing distributions, such as that for weight data, without destroying the identities of individual observations.

Graphs for Quantitative Data - Stem and Leaf Displays

3. Stem and Leaf Displays - Constructing a Display

- **Stems** → represent multiples of 10.
- **Leaves** → represent multiples of 1.
- Arrange a column of numbers, the stems, beginning with 13 (representing the 130s) and ending with 24 (representing the 240s).
- Draw a vertical line to separate the stems and the leaves.
- Enter each raw score into the stem and leaf display for all observations

RAW SCORES					STEM AND LEAF DISPLAY
160	165	135	175		3 5 5
193	168	245	165	13	5
226	169	170	185	14	2 7 1 7 8 0 2 0 2 6 9 8 2 6 4 7 6
152	160	156	154	15	0 3 5 8 9 0 0 0 6 5 5 5
180	170	160	179	16	2 0 0 0 2 5 9
205	150	225	165	17	0 0 5
163	152	190	206	18	0 0 0
157	160	159	165	19	5 6
151	190	172	157	20	
157	150	190	156	21	
220	133	166	135	22	0 5
145	180	158		23	
158	152	152		24	5
172	170	156			

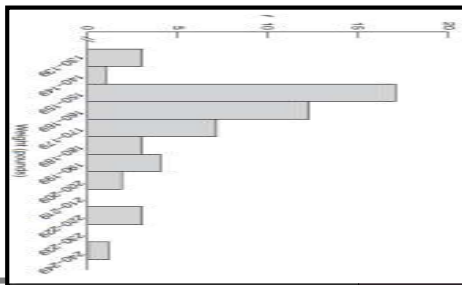
→ raw score of 160 reappears as a leaf of 0 on a stem of 16.



Graphs for Quantitative Data - Stem and Leaf Displays

3. Stem and Leaf Displays

- A glance at the stem and leaf display in the table shows the same pattern of weights depicted by frequency distribution and the histogram.



RAW SCORES					STEM AND LEAF DISPLAY
160	165	135	175		
193	168	245	165	13	3 5 5
226	169	170	185	14	5
152	160	156	154	15	2 7 1 7 8 0 2 0 2 6 9 8 2 6 4 7 6
180	170	160	179	16	0 3 5 8 9 0 0 0 6 5 5 5
205	150	225	165	17	2 0 0 0 2 5 9
163	152	190	206	18	0 0 5
157	160	159	165	19	3 0 0 0
151	190	172	157	20	5 6
157	150	190	156	21	
220	133	166	135	22	6 0 5
145	180	158		23	
158	152	152		24	5
172	170	156			

Graphs for Quantitative Data - Stem and Leaf Displays

3. Stem and Leaf Displays - Selection of Stems

- Depending on the data, the stem with one or more leading digits can be identified.
- Stem value can be in terms of 1, 10, 100, 1000, 0.1, 0.01, etc.
- Eg. an annual income of Rs. 23,784 could be displayed as:

Stem of 23 and a leaf of 784

- Stem and leaf displays represent statistical bargains.
- Produces a description of data that is both clear and complete.
- Even though rarely appearing in published reports, stem and leaf displays often serve as the first step toward organizing data.

Typical Shapes

- Shape of a frequency distribution → Whether expressed as a histogram, a frequency polygon, or a stem and leaf display, an important characteristic of a frequency distribution is its **shape**.
- Some of the more typical shapes for smoothed frequency polygons:
 - Normal shape
 - Bimodal shape
 - Positively skewed
 - Negatively skewed

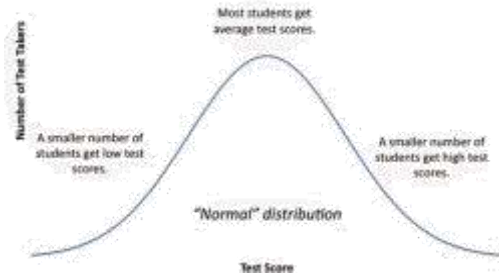
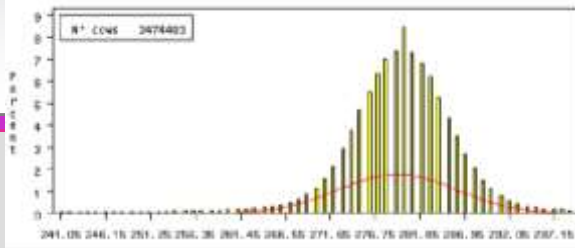
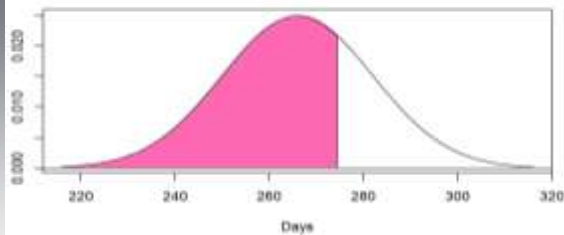
Typical Shapes

Normal

- Any distribution that produces the normal shape can be analyzed with the aid of the **well-documented normal curve**.
- The familiar bell-shaped silhouette of the normal curve can be **superimposed** on many frequency distributions. Few eg.,
 - For uninterrupted gestation periods of human fetuses
 - For scores on standardized tests
 - For the popping times of individual kernels in a batch of popcorn.

Typical Shapes

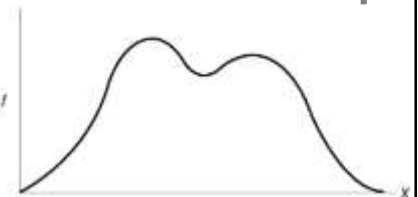
Length of Pregnancy (70th percentile)



Typical Shapes

Bimodal

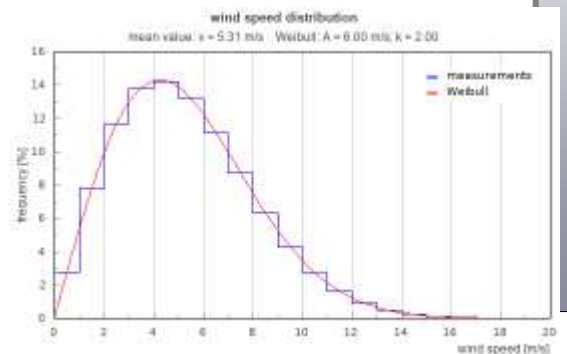
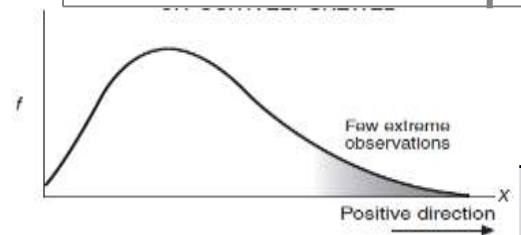
- Any distribution that produces the bimodal shape reflect the coexistence of two different types of observations in the same distribution.
- Eg. 1 The distribution of the ages of residents in a neighborhood consisting largely of either new parents or their infants.
- Eg. 2 The distribution of the Average body weights of humans in a company consisting of males and female employees.



Typical Shapes

Positively Skewed

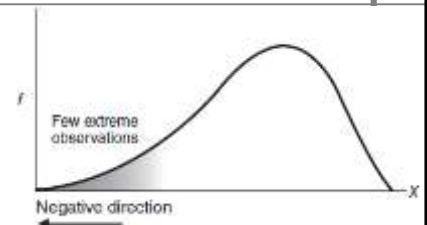
- A lopsided distribution caused by a **few extreme observations** in the **positive direction** (to the right of the majority of observations) is a positively skewed distribution.
- Eg. The distribution of incomes among Indian families has a positive skew → most family incomes under Rs.2,00,000 and relatively very few family incomes spanning a wide range of values above Rs. 200,000 per month.
- Eg. 2 Wind speed distribution



Typical Shapes

Negatively Skewed

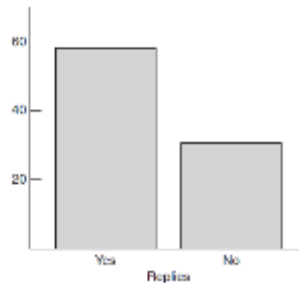
- A lopsided distribution caused by a **few extreme observations** in the **negative direction** (to the left of the majority of observations) is a negatively skewed distributions.
- Eg. Retirement age – more number of retirements after the age of 50.



A GRAPH FOR QUALITATIVE (NOMINAL) DATA

- Bar graph is used.
- Equal segments along the horizontal axis are allocated to the different words or classes that appear in the frequency distribution for qualitative data – equal width for Yes and No.
- Heights of each segment reflect the frequencies for the various words or classes.
- **Gaps** are placed between adjacent bars of bar graphs to emphasize the discontinuous nature of qualitative data.

FACEBOOK PROFILE SURVEY	
Response	<i>f</i>
Yes	56
No	27
Total	83



Constructing Graphs

Step-by-step procedure

1. Decide on the appropriate **type** of graph (histograms and frequency polygons for quantitative data; bar graphs for qualitative data and discrete quantitative data).
2. Draw the **horizontal** axis, then the **vertical** axis (with same width and height respectively).
3. Identify the string of **class intervals** that eventually will be superimposed on the **horizontal** axis.
4. Superimpose the string of class intervals (with gaps for bar graphs) along the entire length of the horizontal axis.
5. Along the entire length of the **vertical** axis, superimpose a progression of convenient numbers.
6. Using the scaled axes, **construct bars** (or dots and lines) to reflect the frequency of observations within each class interval.
7. Supply **labels** for both axes and a **title** for the graph.